

Generalized Probabilistic Topological Field Theories: Summary

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Conceptual skeleton of the GPTFT program: the spacetime axioms, topological field theories as finite algebraic data, the category of GPTs as **Vec** plus positivity, the 1d case as dualizability, and the 2d case as a commutative Frobenius algebra whose finite data are positive GPT data.

THE SPACETIME AXIOMS

- A field theory assigns: $\text{space} \rightarrow \text{system}$, $\text{spacetime} \rightarrow \text{process}$.
- Locality is the compatibility of these assignments:

$$Z(M' \circ M) = Z(M') \circ Z(M), \quad Z(\Sigma \sqcup \Sigma') = Z(\Sigma) \otimes Z(\Sigma').$$

Gluing $\rightarrow$ composition (temporal locality); disjoint union $\rightarrow$ joint system (spatial locality).

- Axioms (Atiyah–Segal [1, 2]): a field theory is a symmetric monoidal functor

$$Z : \mathbf{Bord}_n \longrightarrow \mathcal{C}.$$

- The source is a choice of what spacetime is: in a *topological* theory, bordisms up to diffeomorphism — no metric, no local excitations.
- The target is a choice of what physics is: **Vec**, **Hilb** encode the linear skeleton of quantum theory; a *GPT* target encodes directly the operational, probabilistic content [3–6].
- **GPTFT** := TFT valued in the category of GPTs. Question: *which probabilistic theories admit one?*

TOPOLOGICAL FIELD THEORIES

- $\mathbf{Bord}_n$: objects = closed oriented $(n-1)$ -manifolds; morphisms = compact oriented n -bordisms up to diffeomorphism rel boundary; composition = gluing; $(\sqcup, \emptyset)$ the monoidal structure.
- Diffeomorphism invariance $\Rightarrow$ every manifold decomposes into finitely many elementary pieces $\Rightarrow$ a TFT is *finite algebraic data* in $\mathcal{C}$: values on generators, subject to the relations identifying different decompositions of the same manifold.
- $d = 1$: generators interval, cup, cap; relations the snakes; a theory = a *dualizable object* of $\mathcal{C}$ (cobordism hypothesis [8]).
- $d = 2$: generators disk, pants, copants, reversed disk; a theory = a *commutative Frobenius algebra* in $\mathcal{C}$ [11–13].
- Closed manifolds $\rightarrow$ endomorphisms of **1**, i.e. scalars: the *invariants* of the theory.

THE CATEGORY OF GPTS: VEC PLUS POSITIVITY

Vec (finite-dimensional real): all linear maps, monoidal product $\otimes$, compact closed — duals, ev, coev canonical, snakes automatic. GPT is the same data with one positivity condition per layer:

$$\begin{aligned} V &\rightsquigarrow (V, V^+, u), \\ f &\rightsquigarrow f \text{ with } f(V_A^+) \subseteq V_B^+, \\ V_A \otimes V_B &\rightsquigarrow V_{AB}^+ \in [\otimes_{\min}, \otimes_{\max}], \quad u_A \otimes u_B. \end{aligned}$$

- *Objects*: systems $A = (V_A, V_A^+, u_A)$; V_A^+ a proper cone of unnormalized states, $u_A \in V_A^*$ strictly positive on it. Normalized states $\Omega_A = u_A^{-1}(1) \cap V_A^+$; effects in the dual cone $(V_A^+)^*$; allowed effects $[0, u_A]$; measurements (e_i) , $\sum_i e_i = u_A$. No-restriction by default.

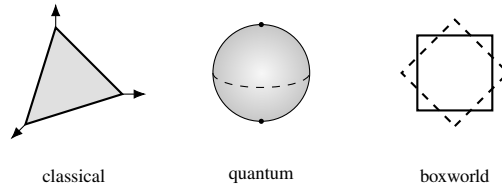


FIG. 1. Normalized-state slices. Classical: simplex (self-dual orthant). Quantum: Bloch ball (self-dual PSD cone). Boxworld: square, dual slice dashed — not self-dual.

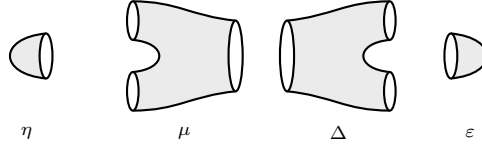


FIG. 2. Generators of $\mathbf{Bord}_2$: disk η , pants μ , copants Δ , reversed disk ε (time flows left to right).

- *Morphisms*: cone-preserving linear maps. $\text{Hom}_{\mathbf{GPT}}(A, B)$ is a *cone*, not a subspace; $\mathbf{GPT}$ is a wide, non-full subcategory of $\mathbf{Vec}$. States and effects are arrows $\mathbf{1} \rightarrow A$, $A \rightarrow \mathbf{1}$; probabilities are composites $e \circ f \circ \omega \in \text{End}(\mathbf{1}) = \mathbb{R}$.
- *Composites*: underlying space $V_A \otimes V_B$ (tomographic locality), unit $u_A \otimes u_B$, composite cone any proper cone in $[\otimes_{\min}, \otimes_{\max}]$. The interval collapses iff one factor is classical [7]; the quantum cone sits strictly between.
- *Departure from Vec*: duals survive on objects ($X^\vee$ swaps state and effect cones onto V^*), but ev and coev are morphisms only if *positive*. Compact closure — automatic in $\mathbf{Vec}$ — becomes a contingent property of the cones.

Examples (Fig. 1): classical $(\mathbb{R}^N, \mathbb{R}_{\geq 0}^N, \sum_i)$, quantum $(H_N(\mathbb{C}), \text{PSD}, \text{tr})$, boxworld (cone over the square [4]). Self-duality of the state cone is the property that will decide everything below.

ONE DIMENSION: DUALIZABILITY

- $\mathbf{Bord}_1$: generated by id_+ , $\text{cup } \cup : \emptyset \rightarrow + \sqcup -$, $\text{cap } \cap : - \sqcup + \rightarrow \emptyset$, modulo the snakes; $S^1 = \cap \circ \sigma \circ \cup$. A 1d theory is a dualizable object $X = Z(+)$ [8].
- In $\mathbf{Vec}$: every object dualizable, $\text{coev}(1) = g := \sum_a f_a \otimes f^a$, snakes = resolution of identity, $Z(S^1) = \dim_{\mathbb{R}} V$. Classification: one integer. Nothing to check.
- In $\mathbf{GPT}$: the dual is *forced*, $X^\vee = (V^*, (V^+)^*, u^\vee)$, and dualizability of $X = (V, V^+, u)$ reduces to two memberships of the *single* element g :

$$\text{cup (state): } g \in V_{X \otimes X^\vee}^+, \quad (1)$$

$$\text{cap (effect): } g \in (V_{X^\vee \otimes X}^+)^*. \quad (2)$$

The cup wants the composite cone large, the cap wants it small; min–max duality exchanges the two ends. Operationally, (1)–(2) assert exact teleportation with g as both resource and measurement [9].

- *Conjecture*: (1)–(2) are satisfiable iff V^+ is *weakly self-dual*. Classical: passes, g separable ($\in \otimes_{\min}$). Quantum: passes, g = maximally entangled element after a transpose twist. Boxworld: no-go (square cone not self-dual; only finitely many twists).
- Koecher–Vinberg [10] + BBLW steering [9]: homogeneous self-dual cones are Euclidean Jordan algebras $\Rightarrow$ theories passing the 1d test are “close to quantum.”

TWO DIMENSIONS: THE FROBENIUS DATA AS GPT DATA

A 2d theory assigns $X := Z(S^1) = (V, V^+, u)$ and one morphism per generator (Fig. 2); the relations (unit, counit, (co)associativity, (co)commutativity, Frobenius) are equalities of linear maps, unchanged from **Vec**. The finite data, read in GPT:

bordism	algebra	GPT positivity
disk $\eta : \emptyset \rightarrow S^1$	unit 1_A	a state: $\eta(1) \in V^+$
rev. disk $\varepsilon : S^1 \rightarrow \emptyset$	counit	an effect: $\varepsilon \in (V^+)^*$
pants $\mu : S^1 \sqcup S^1 \rightarrow S^1$	product	$\mu(V_{XX}^+) \subseteq V^+$
copants $\Delta : S^1 \rightarrow S^1 \sqcup S^1$	coproduct	$\Delta(V^+) \subseteq V_{XX}^+$

Definition (2d GPTFT, finite data). A GPT system $X = (V, V^+, u)$, a composite cone $V_{XX}^+ \in [\otimes_{\min}, \otimes_{\max}]$, and four positive structure maps $\eta, \varepsilon, \mu, \Delta$ as in the table, satisfying the (co)unit, (co)associativity, (co)commutativity and Frobenius relations. Equivalently: a commutative Frobenius algebra in **Vec** all of whose structure maps are positive for the chosen cones.

Conceptual points.

- *Operational reading.* The unit is a preparation (a distinguished unnormalized state created by the disk); the counit is a measurement outcome; merging (μ) and splitting (Δ) of circles are physical transformations.
- *The cone tension returns.* μ is easiest to make positive when V_{XX}^+ is *small* (fewer inputs to control), Δ when it is *large* (more room for outputs): the 2d incarnation of the 1d cup/cap tension, now on genuinely independent data.
- *The 1d problem embeds.* The derived pairing and copairing

$$\beta := \varepsilon \circ \mu \in (V_{XX}^+)^*, \quad \gamma := \Delta(\eta(1)) \in V_{XX}^+$$

are a positive effect and a positive state, and the Frobenius + (co)unit relations force the snake identities for (γ, β) . Hence X is dualizable with $X^\vee \cong X$: the 2d structure contains a solution of the 1d problem *with the dual identified with the object itself*.

- *Self-duality becomes necessary.* Since the dual is forced to be $X^\vee = (V^*, (V^+)^*, u^\vee)$, the isomorphism $X \cong X^\vee$ is a positive linear isomorphism $V \cong V^*$ carrying V^+ onto $(V^+)^*$: *weak self-duality of the state cone is a necessary condition for a 2d GPTFT*. What is conjectural in 1d is forced in 2d. Immediate corollary: boxworld admits no 2d GPTFT.
- *Invariants are nonnegative.* The handle operator $h := \mu \circ \Delta$ is a positive endomorphism of X , and

$$Z(\Sigma_g) = \varepsilon(h^g(\eta(1))) \geq 0 :$$

every closed surface evaluates to a composition of positive data — an (unnormalized) probability. In particular $Z(S^2) = \varepsilon(\eta(1)) \geq 0$ and $Z(T^2) = \dim_{\mathbb{R}} V$.

- *Classical example passes.* $V = \mathbb{R}^N$ with the pointwise algebra: $\eta(1) = \sum_i e_i$, $\varepsilon(e_i) = \lambda_i > 0$, $\mu(e_i \otimes e_j) = \delta_{ij} e_i$, $\Delta(e_i) = \lambda_i^{-1} e_i \otimes e_i$. All four maps are positive (for the unique classical composite cone, $\otimes_{\min} = \otimes_{\max}$), β is nondegenerate: classical 2d GPTFTs exist, one family per choice of weights λ .
- *Quantum is the open case.* The PSD cone passes the necessary self-duality condition, but the obvious products fail: matrix multiplication is neither commutative nor Hermiticity-preserving; the Jordan product is commutative but not associative. Whether $H_N(\mathbb{C})$ ($N \geq 2$) carries *some* commutative, associative, positive Frobenius structure — and more generally, which weakly self-dual cones do — is the 2d classification problem.

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